

Decomposition Rigidity and Compact-State Duality in Relativistic Free-Particle Dynamics

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Abstract We reconstruct classical relativistic free-particle dynamics from a bounded motion–staticity (MS) description, without initially assuming Lorentz transformations, Minkowski chronometry, or a dispersion relation. The primitive state consists of a three-dimensional motion vector and a scalar staticity component, with no prescribed norm on their direct sum. Continuous reversible linear comparison, a bounded nondegenerate pure-staticity orbit, rotational covariance, and independence of axis orientation select an elliptic state shell. A relative channel calibration gives the Euclidean unit shell, whose positive scalar branch describes massive states. An operational radial realization then induces the Lorentzian proper-time relation for pure process segments; a common affine event structure is supplied separately for observer comparison. We require the operational scalar to agree with the canonical generator–displacement response. This dynamical bridge, equivalent to the free proper-time action in a regular Lagrangian description, produces a nonlinear Hamiltonian consistency equation. For finite convex continuously differentiable Hamiltonians, complete representation of the massive MS manifold uniquely selects the standard relativistic square root, without separate isotropy, strict convexity, or rest-value assumptions. Radial and differential rigidity results give complementary conditions, while projection solutions exhibit genuine degeneracies. The positive massive shell has the proper orthochronous Lorentz group as its maximal connected linear covariance group; full observer realization and affine completion require additional assumptions. The construction separates geometric selection, dynamical rigidity, and observer covariance, making their respective inputs explicit.

Keywords: Motion–staticity state, Relativistic free particles, Decomposition rigidity, Hamiltonian rigidity, Convex duality, Lorentz covariance

1 Introduction

The kinematics and dynamics of a relativistic free particle can be organized in several equivalent ways. Standard formulations begin with Lorentzian spacetime geometry, inertial-frame transformations, the relativistic action, or the mass-shell dispersion relation. Reconstruction programs instead ask which weaker structural assumptions are sufficient to recover some or all of these ingredients. Classical examples derive Lorentz kinematics from homogeneity, isotropy, reciprocity, continuity, or composition properties of inertial transformations [1, 2, 3, 4, 5, 6, 7]. The present problem is different and deliberately narrower. We ask whether a bounded vector–scalar description of a massive process can be selected from an operational decomposition principle and, once selected, how strongly compatibility between its operational and canonical representations constrains free-particle dynamics.

The primitive kinematic splitting is

$$V = \mathbb{R}^3 \oplus \mathbb{R}, \quad (1)$$

with spatial rotations acting as $\rho(R) = R \oplus 1$. The three-dimensional sector is interpreted as motion and the one-dimensional sector as staticity. No norm on the direct sum V , no Lorentzian spacetime metric, and no velocity–proper-time relation are assumed at this stage. Here staticity is not introduced as an independent scalar charge. It is intended to represent the complementary non-motion content of one fixed normalized relational state. Accordingly, motion and staticity describe composition rather than overall magnitude: if the motion component could grow without bound while the state remained in the same normalized decomposition sector, motion would acquire an independent extensive scale and the scalar component would cease to have the stated meaning of staticity. This complementarity interpretation motivates the boundedness condition imposed below.

For each unit spatial direction n , the staticity axis and the motion axis along n span a two-dimensional plane. Section 2 models changes of decomposition in this plane by a continuous one-parameter linear comparison group. The orbit of the pure-staticity direction is required to be bounded and to span the plane. In finite dimension this boundedness is independent of the auxiliary norm used to express it. These conditions make the comparison group uniformly bounded; its closure is therefore compact and preserves a positive-definite inner product obtained by the standard compact-group averaging construction [8]. The fixed-state orbit is consequently an ellipse. Spatial rotational covariance makes its eccentricity direction-independent, while independence of the arbitrary orientation assigned to a spatial axis removes the vector–scalar cross term. After one positive relative calibration of the two sectors, the signed normalized shell is Euclidean. Positive scalar orientation selects the massive branch

$$(D, S) \in \mathbb{R}^3 \oplus \mathbb{R}, \quad |D|^2 + S^2 = 1, \quad S > 0. \quad (2)$$

Thus Eq. (2) is not postulated as an independent state-space norm. It is the calibrated physical branch of the bounded reversible comparison structure. The signed completion used in the rigidity argument is auxiliary; no physical interpretation of the negative scalar branch is required.

Closely related velocity–proper-time parametrizations are already known. Brehme used the Euclidean relation between spatial velocity and proper-time rate in relativistic velocity geometry [9]. Savasta and Di Stefano began from the standard velocity–proper-time unit relation and studied associated compact rotations and a related Hamiltonian decomposition (2008, arXiv:0803.4013). The logical direction here is the converse at the first stage: the Euclidean unit shell is not assumed before the comparison structure is specified. Hyperbolic and projective descriptions of relativistic velocity space are likewise standard [10, 11]. Accordingly, neither the unit relation in Eq. (2), its hyperbolic image, nor the usual proper-time formula is claimed as new. The first distinction here is the upstream selection of the Euclidean MS shell from the stated decomposition comparison assumptions rather than its introduction as a geometric ansatz. The second is the inverse dynamical question obtained by requiring the same state to admit operational and canonical representations before the form of the free Hamiltonian has been fixed. Related reconstructions of relativistic energy and momentum from kinematical consistency conditions were developed by Sonogo and Pin [12, 13].

The mathematical ingredients also have established antecedents: compact-group averaging supplies invariant inner products, and the dynamical consistency equation is a multidimensional Clairaut equation associated with Legendre duality [8, 14]. The contribution claimed here is the organization of these ingredients into the stated two-stage reconstruction, including global completeness and differential sufficient conditions and explicit representation degeneracies; neither general method is claimed as new.

The operational construction does not begin by identifying D with v/c using a preexisting coordinate time. A massive process is instead assumed to carry a spatial displacement dx and a positive additive scalar accumulation $d\chi$. In a fixed inertial operational representation the normalized MS state specifies the direction of the vector–scalar increment,

$$\left(\frac{dx}{c}, d\chi\right) = dt(D, S), \quad dt > 0. \quad (3)$$

Here the conversion scale $c > 0$ and the radial coefficient dt arise from the rotationally covariant first-moment readout. At this stage c is not identified with the propagation speed of a particular field, and dt is not imported as an external Newtonian time parameter. Equation (3) defines

$$D = \frac{v}{c}, \quad v := \frac{dx}{dt}, \quad S = \frac{d\chi}{dt}. \quad (4)$$

Combining this realization with Eq. (2) gives

$$c^2 d\chi^2 = c^2 dt^2 - dx^2. \quad (5)$$

Only after this Lorentzian quadratic relation has been induced do we define $d\tau := d\chi$. Hence

$$S = \frac{d\tau}{dt},$$

and the primitive scalar accumulation acquires its standard proper-time interpretation. This construction does not derive temporal ordering or the existence of intrinsic accumulation from the MS decomposition alone; it derives the relation among intrinsic accumulation, observer-assigned duration, and spatial displacement once those process data are supplied.

The same normalized state is assigned an independent dimensional generator scale. For $Q > 0$, define

$$(d, q) = Q(D, S). \quad (6)$$

The radial quantities dt and Q have distinct roles: dt is the observer-assigned extent of a process segment, whereas Q is an intrinsic scale of the massive generator. In particular,

$$q = QS = Q \frac{d\tau}{dt}.$$

The second rigidity problem begins when the same MS state is required to have a canonical representation. Let $H(p)$ be an initially unspecified differentiable autonomous free Hamiltonian. Hamilton's equation gives $v = \nabla_p H$. Introduce

$$P_H = \left(\frac{H}{c}, p\right), \quad \Pi_H = P_H^\flat = \left(-\frac{H}{c}, p\right), \quad (7)$$

where the metric dual is taken with respect to the Lorentzian form already induced by the operational construction. For $dX = (c dt, dx)$, the canonical generator–translation pairing is

$$\Pi_H(dX) = p \cdot dx - H dt.$$

Along a Hamiltonian trajectory this becomes

$$\Pi_H(dX) = -(H - p \cdot \nabla H) dt.$$

Requiring the operational and Hamiltonian descriptions to represent the same scalar MS component gives

$$q = Q \frac{d\tau}{dt} = \frac{H - p \cdot \nabla H}{c}, \quad (8)$$

$$q dt = Q d\tau = -\frac{1}{c} \Pi_H(dX). \quad (9)$$

If a regular Lagrangian description exists, ordinary Legendre duality also gives $q = -L/c = (H - p \cdot v)/c$. The individual Legendre and proper-time identities are standard; the representation requirement is that they describe the same scalar component before H is known.

The complete common representation is therefore

$$(D, S) = \left(\frac{v}{c}, \frac{d\tau}{dt} \right) = \left(\frac{\nabla H}{c}, \frac{H - p \cdot \nabla H}{cQ} \right). \quad (10)$$

The compact normalization then gives the nonlinear consistency equation

$$\frac{|\nabla H|^2}{c^2} + \frac{(H - p \cdot \nabla H)^2}{c^2 Q^2} = 1, \quad H - p \cdot \nabla H > 0. \quad (11)$$

This equation is not postulated as a dispersion relation. It is the compatibility condition obtained by representing one already selected MS state in operational and canonical variables.

Our strongest dynamical result uses canonical representation completeness. For a finite convex C^1 Hamiltonian, assume Eq. (11) globally and require the canonical representation to realize the complete massive MS manifold. Equivalently,

$$\nabla H(\mathbb{R}^3) = \{v : |v| < c\}. \quad (12)$$

Then

$$H(p) = c \sqrt{|p|^2 + Q^2}. \quad (13)$$

No independent isotropy assumption on H , strict convexity, or Hessian nondegeneracy is needed. Complementary results show that radial C^2 solutions reduce to the constant branch or Eq. (13), while dense Hessian nondegeneracy again forces the square-root branch without assuming radiality. The additional hypotheses are substantive: for a proper orthogonal projection R ,

$$H_R(p) = c \sqrt{Q^2 + |Rp|^2} \quad (14)$$

satisfies the same pointwise consistency equation but realizes only a lower-dimensional motion sector. Thus the master equation alone does not ensure a faithful representation of all canonical directions.

The forward problem provides a useful separation between the two rigidity steps. For an arbitrary compact convex state body K with support function h_K , standard Legendre–Fenchel duality gives

$$H_{K,Q}(p) = c h_K(p, Q). \quad (15)$$

Hence compactness by itself does not select the relativistic square root, and Hamiltonian consistency does not select the Euclidean state geometry from an arbitrary compact body [15, 16]. Appendix B exhibits an L^r -type family and shows separately that the conventional Newtonian quadratic limit fixes only local rest geometry in the general smooth case. In the present construction the Euclidean shell is

instead selected upstream by the bounded decomposition-comparison assumptions of Sec. 2; Sec. 3 then establishes dynamical rigidity on that state manifold.

For the nondegenerate branch, writing $E := H(p)$ and $P = (E/c, p)$ gives

$$\eta(P, P) = -Q^2, \quad \eta = \text{diag}(-1, +1, +1, +1). \quad (16)$$

With $Q = mc$, this is the standard relation $E^2 - c^2|p|^2 = m^2c^4$. The compact completion $S \geq 0$ is projectively equivalent to the future causal rays: $S > 0$ gives the timelike sector and $S = 0$ the null boundary. A finite nonzero null momentum is obtained by the scale-free limit $Q \rightarrow 0^+$ at fixed nonzero momentum, rather than by the infinite-momentum limit on a fixed massive shell.

No Lorentz transformation law is assumed in either rigidity step. Observer covariance is introduced only after the Lorentzian translation form and the positive massive shell have been obtained. Operational translations form a vector space $\mathfrak{t}$, with canonical generator $\Pi \in \mathfrak{t}^*$. A continuous change of inertial description that preserves process addition acts linearly on $\mathfrak{t}$, while invariance of the scalar pairing $\Pi(dX)$ induces the dual action on generators. Preservation of the positive massive shell restricts the connected linear covariance to $\text{SO}^+(1, 3)$. Under comoving-frame completeness and unrestricted spatial reorientation the full proper orthochronous Lorentz group is realized, and allowing changes of event origin gives the proper orthochronous Poincaré group.

The scope is correspondingly explicit. The construction assumes a three-dimensional Euclidean spatial readout and its rotational organization, a primitive vector–scalar motion–staticity splitting, continuous linear decomposition comparison with a bounded spanning pure-staticity orbit, spatial rotational covariance, independence of the arbitrary orientation of each spatial axis, positive additive intrinsic process accumulation, and the first-moment assumptions used in the operational cone construction. It does not derive spatial dimension, temporal ordering, the vector–scalar split, or the comparison principles from a structureless set of events. Its claim is two-stage: these inputs first select the Euclidean compact MS geometry up to channel calibration; compatibility of that state with a complete or sufficiently nondegenerate canonical representation then selects the standard relativistic free Hamiltonian and its Lorentz–Poincaré covariance.

The paper is organized as follows. Section 2 proves the decomposition-rigidity result, constructs the operational radial realization, and develops the projective causal completion. Section 3 establishes the operational–canonical bridge, convex-dual formulation, and Hamiltonian rigidity results. Section 4 derives Lorentz and Poincaré covariance from the translation–generator structure. Section 5 summarizes the assumptions, logical status, and extensions. The Appendixes provide the operational linear-extension argument, convex-dual examples and Newtonian rest-geometry analysis, and explicit degenerate projection solutions.

2 Compact State, Operational Duration, and Causal Completion

The primitive motion–staticity (MS) description is intended to encode the relative organization of motion and staticity without assuming a Lorentzian spacetime metric or an external coordinate time. We assume a three-dimensional Euclidean spatial readout and a class of inertial massive processes that can be described relationally by an observer. No absolute spatial rest frame or independently observable observer self-velocity is introduced.

A massive process is further assumed to carry an additive scalar accumulation, denoted initially by χ . Process concatenation is taken to have a chosen positive orientation, and for every nontrivial

positively oriented segment,

$$d\chi > 0. \quad (17)$$

The increment $d\chi$ is taken to be attached to the massive process segment itself rather than to a particular inertial description; observer dependence enters through the spatial readout, the radial duration, and the corresponding motion–staticity decomposition. At this stage χ is only an extensive scalar process variable. No metric interpretation as proper time or observer duration is assigned to it.

2.1 Compact state and operational radial realization

Let

$$V = \mathbb{R}^3 \oplus \mathbb{R} \quad (18)$$

carry the spatial-rotation action

$$\rho(R) = R \oplus 1, \quad R \in \text{SO}(3). \quad (19)$$

Thus the first summand transforms as a spatial vector and the second as a spatial scalar. This vector–scalar splitting is taken as primitive; no norm on the direct sum V is assumed at this stage.

The term “staticity” carries a specific complementarity meaning in this construction. The scalar sector is not an additional independent amount that can coexist with an arbitrarily amplified motion sector; rather, it is the non-motion component complementary to motion within one fixed normalized relational state. The normalized MS variables are therefore intended to describe how a fixed state content is distributed between motion and staticity, not how large that content is. An independent overall scale will be introduced only later through the radial quantities.

This interpretation already constrains the admissible comparison structure. If repeated changes of motion–staticity decomposition could drive the motion component arbitrarily far while remaining within the same normalized state sector, then the motion coordinate would itself carry an independent extensive magnitude. The scalar component would then no longer quantify what remains non-motional in that same state, and the intended motion–staticity complementarity would be lost. We therefore require the fixed-state decomposition orbit to remain bounded. The condition does not assume a quadratic norm or a particular compact geometry; it only excludes unbounded gain within a sector whose variables are supposed to encode composition rather than magnitude.

For a unit spatial direction $n \in \mathbb{S}^2$, set

$$e_n := (n, 0), \quad s := (0, 1), \quad W_n := \text{span}\{e_n, s\}. \quad (20)$$

The plane W_n contains the staticity axis and the motion axis along the oriented spatial direction n .

Assumption 2.1 (Bounded reversible decomposition comparison). *For every $n \in \mathbb{S}^2$, changes of motion–staticity decomposition along the corresponding spatial axis admit a formal signed comparison completion*

$$T_n : (\mathbb{R}, +) \longrightarrow \text{GL}(W_n) \quad (21)$$

with the following properties.

(i) T_n is continuous, $T_n(0) = I$, and

$$T_n(t_1 + t_2) = T_n(t_1)T_n(t_2). \quad (22)$$

The real-linear action is part of the comparison principle: comparison is required to respect additive and homogeneous composition in the ambient linear completion of unnormalized state contributions. No linear structure is thereby imposed on the normalized orbit itself.

(ii) *The pure-staticity orbit*

$$O_n := \{T_n(t)s : t \in \mathbb{R}\} \quad (23)$$

is bounded and spans W_n . Physically, boundedness expresses the complementarity just described: changing the decomposition of a fixed normalized state may redistribute motion and staticity but may not generate an independent unbounded scale in the motion sector. Without this condition, the scalar coordinate could no longer be interpreted as the complementary staticity content of the same normalized state. Boundedness refers only to the finite-dimensional vector-space topology: equivalently, the orbit is bounded with respect to any auxiliary norm on W_n . No such auxiliary norm is assigned physical significance.

(iii) *The comparison is rotationally covariant. For every $R \in \text{SO}(3)$,*

$$T_{Rn}(t)\rho(R)|_{W_n} = \rho(R)|_{W_n}T_n(t). \quad (24)$$

(iv) *The set of allowed decompositions associated with a spatial axis does not depend on which orientation is assigned to that axis. Under the canonical identification $W_{-n} = W_n$,*

$$O_{-n} = O_n. \quad (25)$$

Assumption 2.1 does not postulate a quadratic norm. Its geometric content is specified by the following structural requirements: fixed-state decomposition comparison is continuous, reversible, composition-compatible, bounded and nondegenerate, and it respects the rotational organization of the three-dimensional motion sector without depending on an arbitrary choice of orientation for a spatial axis.

Theorem 2.2 (Rigidity of bounded motion–staticity decomposition). *Under Assumption 2.1, there exists a constant $\kappa > 0$, independent of n , such that*

$$O_n = \{xe_n + ys : \kappa^2 x^2 + y^2 = 1\}. \quad (26)$$

Consequently the union of the directional orbits is the rotational ellipsoid

$$\Sigma_{\text{raw}} = \{(z, \sigma) \in \mathbb{R}^3 \oplus \mathbb{R} : \kappa^2 |z|^2 + \sigma^2 = 1\}. \quad (27)$$

After the positive relative calibration

$$D := \kappa z, \quad S := \sigma, \quad (28)$$

the signed normalized shell takes the canonical Euclidean form

$$\Sigma = \{(D, S) : |D|^2 + S^2 = 1\}. \quad (29)$$

Proof. Fix n and abbreviate $W = W_n$, $T(t) = T_n(t)$, and $O = O_n$. Choose any auxiliary norm $\|\cdot\|_0$ on W . By boundedness there is $M < \infty$ such that

$$\|T(t)s\|_0 \leq M \quad \text{for all } t \in \mathbb{R}. \quad (30)$$

Because O spans the two-dimensional space W , there is t_0 for which s and $u := T(t_0)s$ are linearly independent. Hence every $w \in W$ can be written uniquely as $w = as + bu$. The group law gives

$$T(t)w = aT(t)s + bT(t+t_0)s. \quad (31)$$

Therefore $\|T(t)w\|_0 \leq M(|a| + |b|)$. Since all norms on a finite-dimensional space are equivalent, there is $C > 0$ such that $|a| + |b| \leq C\|w\|_0$. Thus

$$\sup_{t \in \mathbb{R}} \|T(t)\|_{\text{op}} < \infty. \quad (32)$$

The same estimate includes $T(t)^{-1} = T(-t)$.

Let

$$K := \overline{\{T(t) : t \in \mathbb{R}\}}, \quad (33)$$

where the closure is taken in $\text{End}(W)$. In finite dimension the uniform bound makes this closure compact. Moreover every limit point is invertible: if $T(t_j) \rightarrow A$, the uniformly bounded sequence $T(-t_j)$ has a convergent subsequence, say to B , and then $AB = BA = I$. Hence K is a compact subgroup of $\text{GL}(W)$.

Choose any auxiliary inner product $\langle \cdot, \cdot \rangle_0$ on W and let μ be normalized Haar measure on K . Define

$$B_n(v, w) := \int_K \langle gv, gw \rangle_0 d\mu(g). \quad (34)$$

Then B_n is a positive-definite inner product and is invariant under K , hence under every $T(t)$. Consequently

$$B_n(T(t)s, T(t)s) = B_n(s, s). \quad (35)$$

Thus O lies on an ellipse. In a B_n -orthonormal basis, the connected one-parameter group $T(\mathbb{R})$ lies in $\text{SO}(2)$. It is nontrivial because O spans W , so for some $\omega_n \neq 0$ it has the form of rotations through angle $\omega_n t$. Hence the orbit is the entire ellipse

$$O = \{v \in W : B_n(v, v) = B_n(s, s)\}. \quad (36)$$

Write, in the ordered basis (e_n, s) ,

$$B_n(xe_n + ys, xe_n + ys) = a_n x^2 + 2c_n xy + b_n y^2, \quad (37)$$

with $a_n > 0$, $b_n > 0$, and $a_n b_n - c_n^2 > 0$. Choose $R_\pi \in \text{SO}(3)$ with $R_\pi n = -n$. Rotational covariance gives $O_{-n} = \rho(R_\pi)O_n$, while axis-orientation independence gives $O_{-n} = O_n$. Therefore the ellipse is invariant under $(x, y) \mapsto (-x, y)$. Since the orbit is a nondegenerate full ellipse, this invariance forces $c_n = 0$. Dividing the level-set equation by $b_n = B_n(s, s)$ yields

$$\kappa_n^2 x^2 + y^2 = 1, \quad \kappa_n^2 := \frac{a_n}{b_n} > 0. \quad (38)$$

For any $m \in \mathbb{S}^2$, choose $R \in \text{SO}(3)$ with $Rn = m$. Rotational covariance maps e_n to e_m , fixes s , and maps O_n onto O_m . Consequently $\kappa_m = \kappa_n$. Denoting the common positive constant by κ proves Eq. (26); taking the union over directions gives Eq. (27), and the relative calibration (28) gives Eq. (29). $\square$

Remark 2.3 (Logical status of the Euclidean shell). *The proof uses an auxiliary norm only to express topological boundedness and an auxiliary inner product only inside the compact-group averaging argument. Neither is part of the physical state geometry. The positive-definite quadratic form preserved by the comparison group is obtained as a consequence of bounded reversible linear comparison. The numerical equality of the motion and staticity coefficients in Eq. (29) is then a relative channel calibration; the invariant content before that calibration is the elliptic quadratic class together with rotational isotropy of the motion sector.*

The group completion in Assumption 2.1 is formal and signed; the full comparison orbit contains both signs of the scalar coordinate. We now choose the scalar orientation so that positively oriented massive processes occupy the branch $S > 0$. This orientation convention is calibrated below by $d\chi > 0$ and the positive scalar readout coefficient. The set $S = 0$ is retained as the boundary of the compact completion, while no physical interpretation of $S < 0$ is required.

The normalized massive state manifold and its compact completion are

$$S := \{(D, S) : |D|^2 + S^2 = 1, S > 0\}, \quad (39)$$

$$\bar{S} := \{(D, S) : |D|^2 + S^2 = 1, S \geq 0\}. \quad (40)$$

For a massive system, let $Q > 0$ denote the fixed overall generator scale and define the dimensional generator channels by

$$(d, q) := Q(D, S). \quad (41)$$

Equivalently,

$$D = \frac{d}{Q}, \quad S = \frac{q}{Q}. \quad (42)$$

The dimensional shell is therefore

$$|d|^2 + q^2 = Q^2, \quad q > 0, \quad (43)$$

and its associated positive-definite quadratic magnitude is

$$|d|^2 + q^2. \quad (44)$$

This is an internal state-space structure obtained only after the decomposition-rigidity step and channel calibration; it is not an assumed spacetime metric.

To separate normalized state direction from extensive process magnitude, let

$$C := \text{cone}(S) \subset V \quad (45)$$

denote the positive cone generated by the massive states. The equivalent barycentric construction and its linear extension are given in Appendix A.

The operational first-moment readout is taken to have the form

$$R : C \longrightarrow \mathbb{R}_{\text{disp}}^3 \oplus \mathbb{R}_\chi. \quad (46)$$

We impose barycentric sufficiency: preparations with the same unnormalized barycentric state have the same first-moment readout. We also assume additivity of this readout under combination of preparations and positive homogeneity under common positive rescaling. These assumptions make R a well-defined additive and positively homogeneous cone map, which extends uniquely to a real-linear map

$$\tilde{R} : V \longrightarrow \mathbb{R}_{\text{disp}}^3 \oplus \mathbb{R}_\chi. \quad (47)$$

No linear structure is thereby imposed directly on the normalized pure-state manifold.

Spatial covariance requires

$$\tilde{R}(RD, S) = (R \oplus 1)\tilde{R}(D, S). \quad (48)$$

Since the vector and scalar sectors are inequivalent irreducible $\text{SO}(3)$ representations,

$$\tilde{R}(D, S) = (\alpha D, \beta S) \quad (49)$$

for constants α, β . Positive process orientation fixes $\beta > 0$; nondegeneracy and orientation of the spatial channel allow us to take $\alpha > 0$.

We now work in one fixed inertial operational representation and suppress its observer label. Neither the normalized MS direction nor its positive radial coefficient is assumed to be observer invariant. A different inertial observer may assign both a different direction and a different radial coefficient to the same physical process segment; their relation will be determined only after the observer covariance structure has been constructed.

For a pure massive segment,

$$Y = \lambda(D, S), \quad \lambda > 0, \quad (50)$$

the readout is

$$(dx, d\chi) = \lambda(\alpha D, \beta S). \quad (51)$$

The coefficient λ parametrizes the extensive size of the segment within the chosen operational representation.

Define the positive conversion scale and observer duration by

$$c := \frac{\alpha}{\beta}, \quad dt := \beta\lambda > 0. \quad (52)$$

At this stage c is the common conversion convention used to express the spatial and scalar readout channels of the inertial operational representations in common units. This is a calibration convention, not yet a consequence of Lorentz covariance, and no identification with the propagation speed of a particular field is assumed. Its causal role is established only after the observer covariance structure is derived; Sec. 4 separately gives conditions under which the same conversion scale is universal across massive species.

Equation (51) becomes

$$\left(\frac{dx}{c}, d\chi\right) = dt(D, S). \quad (53)$$

The normalized pair (D, S) therefore carries directional information, whereas dt is the extensive radial coefficient assigned by the observer.

Using $|D|^2 + S^2 = 1$,

$$dt = \sqrt{d\chi^2 + \frac{|dx|^2}{c^2}}. \quad (54)$$

For successive segments of the same inertial process, the common MS direction is fixed and the positive radial coefficients add. Thus t is an affine process parameter, unique up to an additive origin.

This assertion concerns successive segments with the same MS direction. It does not identify addition of barycentric state data with addition of spacetime translations. For different directions, the Euclidean radial coefficient is generally not additive. For example, if $X_{\pm} = (\pm ae_1, \sqrt{1-a^2})$, $0 < a < 1$, then $\|X_+ + X_-\|_2 = 2\sqrt{1-a^2} < 2 = \|X_+\|_2 + \|X_-\|_2$. For a concatenated history, duration is the sum of the segment durations; Eq. (54) is not reapplied to the total displacement and accumulated scalar as though they formed one pure segment. The common event and translation structure used in Sec. 4 is an additional assumption.

We may now define

$$v := \frac{dx}{dt}, \quad (55)$$

so that

$$D = \frac{v}{c}, \quad S = \frac{d\chi}{dt}. \quad (56)$$

For the pure-staticity state $(D, S) = (0, 1)$, Eq. (53) gives

$$dx = 0, \quad dt = d\chi. \quad (57)$$

Thus the static state calibrates the scalar accumulation against the observer duration.

We assume operational completeness: every state in $\mathcal{S}$ is realized by an inertial massive process. Consequently the realized observer velocities fill the open ball

$$B_c := \{v \in \mathbb{R}^3 : |v| < c\}. \quad (58)$$

2.2 Lorentzian chronometry and projective causal completion

The compact-state relation and Eq. (56) imply

$$\frac{|v|^2}{c^2} + \left(\frac{d\chi}{dt}\right)^2 = 1, \quad (59)$$

or equivalently

$$c^2 d\chi^2 = c^2 dt^2 - dx^2. \quad (60)$$

Relations of this form are standard in relativistic velocity–proper-time parametrizations [9]. In particular, the Euclidean unit relation between spatial velocity and proper-time rate is not claimed as new here. The role specific to the present construction is that observer duration has first been introduced as the radial coefficient of the operational MS realization rather than as an external coordinate-time parameter.

Introduce

$$dX := (c dt, dx) \in \mathbb{R}^4 \quad (61)$$

and define

$$\eta := \text{diag}(-1, +1, +1, +1). \quad (62)$$

Equation (60) then reads

$$\eta(dX, dX) = -c^2 d\chi^2. \quad (63)$$

By operational completeness, the realized values of D fill the open unit ball, so the corresponding increments dX fill an open future-timelike cone in the translation space. Hence the operational relation determines the quadratic form uniquely: if another quadratic form agreed with Eq. (63) on all realized increments, the difference of the two quadratic forms would vanish on an open set and therefore vanish identically. Thus the operational radial relation uniquely induces the Lorentzian quadratic form on the four-dimensional translation coordinates. At this stage, however, invariance under changes of inertial observer has not been assumed or proved.

We now identify the intrinsic scalar increment with proper time by defining

$$d\tau := d\chi. \quad (64)$$

Hence τ and χ differ at most by an additive choice of origin, and

$$S = \frac{d\tau}{dt} = \sqrt{1 - \frac{|v|^2}{c^2}}. \quad (65)$$

The Euclidean norm $|D|^2 + S^2 = 1$ and the Lorentzian norm $\eta(dX, dX) = -c^2 d\tau^2$ therefore belong to different spaces: the former to the compact internal state geometry, the latter to operational translation space.

The compact state has the standard hyperbolic representation [9, 10, 11]. Let

$$\mathcal{H}^+ := \{U \in \mathbb{R}^4 : \eta(U, U) = -1, U^0 > 0\}. \quad (66)$$

For $S > 0$, define

$$\Phi : \mathcal{S} \longrightarrow \mathcal{H}^+, \quad \Phi(D, S) = \left(\frac{1}{S}, \frac{D}{S} \right). \quad (67)$$

Since

$$\eta(\Phi, \Phi) = \frac{-1 + |D|^2}{S^2} = -1,$$

Φ maps $\mathcal{S}$ into the future unit hyperboloid. Its inverse is

$$\Phi^{-1}(U^0, U) = \left(\frac{U}{U^0}, \frac{1}{U^0} \right), \quad (68)$$

so the map is a diffeomorphism.

Its operational meaning follows directly from Eq. (53):

$$\Phi(D, S) = \left(\frac{dt}{d\tau}, \frac{1}{c} \frac{dx}{d\tau} \right). \quad (69)$$

After the Lorentzian interpretation has been made, this is the standard dimensionless future four-velocity. The four-velocity formula itself is standard; here Φ also records how the compact state is lifted by normalization with respect to intrinsic massive accumulation.

The compact boundary is most naturally described projectively. Let

$$C^+ := \{Y \in \mathbb{R}^4 \setminus \{0\} : Y^0 > 0, \eta(Y, Y) \leq 0\}, \quad (70)$$

and let $\mathbb{P}_+(C^+)$ denote the quotient by positive rescaling. Then

$$\widehat{\Phi} : \overline{\mathcal{S}} \longrightarrow \mathbb{P}_+(C^+), \quad \widehat{\Phi}(D, S) = [(1, D)] \quad (71)$$

is a bijection. Indeed,

$$\eta((1, D), (1, D)) = -1 + |D|^2 = -S^2. \quad (72)$$

Thus $S > 0$ corresponds to future-timelike rays, whereas

$$S = 0, \quad |D| = 1 \quad (73)$$

corresponds to future-null rays. Conversely, every future causal ray has a unique representative $(1, D)$ with $|D| \leq 1$, and the corresponding compact-state coordinate is $S = \sqrt{1 - |D|^2}$. Hence

$$\overline{\mathcal{S}} \simeq \mathbb{P}_+(C^+) \quad (74)$$

as compact projective state spaces.

For $S > 0$, Φ selects the unique future representative of Lorentzian norm -1 . At the null boundary this normalization ceases to exist, and

$$\frac{dt}{d\tau} = \frac{1}{S} \longrightarrow +\infty \quad (S \rightarrow 0^+). \quad (75)$$

The divergence is therefore a normalization effect, not a singularity of the projective causal ray.

Finally, the boundary $S = 0$ should not be produced by simply setting $Q = 0$ in $(D, S) = (d, q)/Q$, which is defined only for $Q > 0$. The finite-momentum null completion of the massive Hamiltonian family is described in Sec. 3.

The temporal hierarchy used below is therefore limited and explicit. The additive variable χ is a primitive scalar process accumulation; the MS direction determines how that scalar channel combines with spatial displacement; the positive radial coefficient defines an observer-dependent duration; and only then is $d\chi$ identified with the proper-time increment $d\tau$ of the induced Lorentzian geometry.

3 Compact-State Duality and Hamiltonian Rigidity

For a massive state,

$$(d, q) = Q(D, S), \quad Q > 0, \quad (76)$$

whereas the operational realization of Sec. 2 gives

$$S = \frac{d\tau}{dt}, \quad D = \frac{v}{c}.$$

Thus

$$q = QS = Q \frac{d\tau}{dt}. \quad (77)$$

The purpose of this section is to determine when the same compact state can be represented by an initially unspecified autonomous free Hamiltonian. No relativistic dispersion relation or four-momentum transformation law is assumed.

3.1 Canonical response and compact convex duality

Let

$$H : \mathbb{R}^3 \longrightarrow \mathbb{R} \quad (78)$$

be differentiable. Hamilton's equation gives

$$v = \nabla_p H. \quad (79)$$

Using the Lorentzian form induced in Sec. 2, introduce the generator vector candidate and its metric-dual action covector,

$$P_H := \left(\frac{H}{c}, p \right), \quad \Pi_H := P_H^\flat = \left(-\frac{H}{c}, p \right). \quad (80)$$

For

$$dX = (c \, dt, dx),$$

the canonical pairing is

$$\Pi_H(dX) = p \cdot dx - H \, dt. \quad (81)$$

Along the Hamiltonian trajectory $dx = \nabla H dt$,

$$\Pi_H(dX) = -(H - p \cdot \nabla H)dt. \quad (82)$$

This defines the scalar canonical candidate

$$q_H := -\frac{\Pi_H(dX)}{c dt} = \frac{H - p \cdot \nabla H}{c}. \quad (83)$$

If a regular Lagrangian description exists, ordinary Legendre duality gives

$$L = p \cdot v - H, \quad v = \nabla H. \quad (84)$$

The MS representation-consistency requirement is

$$D_H = D, \quad q_H = q.$$

Consequently the scalar component has the equivalent representations

$$q = QS = Q \frac{d\tau}{dt} = -\frac{L}{c} = \frac{H - p \cdot v}{c} = \frac{H - p \cdot \nabla H}{c}, \quad (85)$$

$$q dt = Q d\tau = -\frac{1}{c} \Pi_H(dX). \quad (86)$$

These two relations are the scalar representation bridge used throughout the remainder of the construction. Their individual Legendre and proper-time identities are standard; related velocity–proper-time decompositions were discussed, for example, by Savasta and Di Stefano in the preprint cited in Sec. 1. The additional requirement here is that the operational, Lagrangian, and Hamiltonian quantities represent the same MS scalar component before the form of H has been fixed.

This identification is a substantive dynamical assumption. When a regular Lagrangian description exists, it is equivalent to $L dt = -cQ d\tau$, the usual free-particle proper-time action density with $Q = mc$. Thus the reconstruction makes the dynamical bridge explicit; it does not obtain that bridge from compact state geometry alone.

The vector and scalar components can therefore be written together as

$$(D, S) = \left(\frac{v}{c}, \frac{d\tau}{dt} \right) = \left(\frac{\nabla H}{c}, \frac{H - p \cdot \nabla H}{cQ} \right). \quad (87)$$

The inverse problem of this paper is precisely the problem of determining which free Hamiltonians can realize Eq. (87) globally.

Imposing the compact-state normalization gives

$$\frac{|\nabla H|^2}{c^2} + \frac{(H - p \cdot \nabla H)^2}{c^2 Q^2} = 1, \quad (88)$$

with the massive branch condition

$$H - p \cdot \nabla H > 0. \quad (89)$$

Equivalently, Eq. (86) gives the generator–displacement relation

$$-\Pi_H(dX) = cQ d\tau. \quad (90)$$

The master equation is therefore a compact-state consistency condition, not an independent dispersion postulate.

On the positive branch it can equivalently be written

$$H = p \cdot \nabla H + cQ \sqrt{1 - |\nabla H|^2/c^2}.$$

This is a multidimensional Clairaut equation of the form $H = p \cdot \nabla H - L_Q(\nabla H)$. Its relation to Legendre duality and to envelope and degenerate solutions is standard [14]. The results below specify global hypotheses under which this particular equation admits only the complete massive branch.

For comparison, the forward construction is standard convex duality. Let $K \subset \mathbb{R}^3 \oplus [0, \infty)$ be nonempty, compact, and convex, with support function

$$h_K(\xi, \zeta) := \sup_{(D, S) \in K} (\xi \cdot D + \zeta S). \quad (91)$$

Write $\Omega = \text{proj}_D K$ and $\sigma_K(D) = \max\{S : (D, S) \in K\}$. For $Q > 0$, define the extended-real convex Lagrangian

$$L_{K, Q}(v) = \begin{cases} -cQ \sigma_K(v/c), & v/c \in \Omega, \\ +\infty, & v/c \notin \Omega. \end{cases}$$

Compactness and convexity of K make σ_K upper semicontinuous and concave on Ω . Its Legendre–Fenchel conjugate is

$$L_{K, Q}^*(p) = c \sup_{D \in \Omega} (p \cdot D + Q \sigma_K(D)).$$

The corresponding Legendre–Fenchel construction gives

$$H_{K, Q}(p) = c h_K(p, Q), \quad (92)$$

a standard support-function identity [15, 16]. Details and non-Euclidean examples are collected in Appendix B.

For the Euclidean state body

$$K_2^+ := \text{conv } \mathcal{S} = \{(D, S) : |D|^2 + S^2 \leq 1, S \geq 0\}, \quad (93)$$

one obtains

$$h_{K_2^+}(p, Q) = \sqrt{|p|^2 + Q^2}, \quad H_Q(p) = c \sqrt{|p|^2 + Q^2}. \quad (94)$$

The associated closed convex Lagrangian is

$$L_Q(v) = \begin{cases} -cQ \sqrt{1 - |v|^2/c^2}, & |v| \leq c, \\ +\infty, & |v| > c. \end{cases} \quad (95)$$

For $Q > 0$, the boundary value $|v| = c$ belongs to the closed convex extension and is not a finite-momentum massive state.

This forward construction also shows that compactness alone does not select the relativistic square root: different compact bodies generally have different support functions. In the present construction the Euclidean state body is selected upstream by the decomposition-rigidity result of Theorem 2.2, rather than by compactness alone. The Hamiltonian rigidity question is the converse one: once this state geometry is fixed, how strongly does common operational–canonical representation constrain the free dynamics?

3.2 Hamiltonian rigidity and causal completion

Definition 3.1 (Canonical representation completeness). *A Hamiltonian satisfying Eqs. (88) and (89) is representation-complete if*

$$\Psi_H(p) := \left(\frac{\nabla H(p)}{c}, \frac{H(p) - p \cdot \nabla H(p)}{cQ} \right) \quad (96)$$

maps $\mathbb{R}^3$ onto $\mathcal{S}$.

Because the positive scalar component is fixed by the vector component, representation completeness is equivalent to

$$\nabla H(\mathbb{R}^3) = B_c = \{v : |v| < c\}. \quad (97)$$

Theorem 3.2 (Rigidity from complete MS representation). *Let $H : \mathbb{R}^3 \rightarrow \mathbb{R}$ be finite, convex, and C^1 . Suppose*

$$c > 0, \quad Q > 0, \quad (98)$$

Eqs. (88) and (89) hold globally, and H is representation-complete. Then

$$H(p) = c \sqrt{|p|^2 + Q^2}. \quad (99)$$

Proof. Let

$$H^*(v) := \sup_p (p \cdot v - H(p)) \quad (100)$$

be the convex conjugate. For $|v| < c$, representation completeness gives a momentum with $v = \nabla H(p)$. Fenchel equality and the positive branch of Eq. (88) then give

$$H^*(v) = -cQ \sqrt{1 - \frac{|v|^2}{c^2}}, \quad |v| < c. \quad (101)$$

The positive branch also implies $|\nabla H| < c$. Integration along line segments yields

$$|H(p) - H(q)| \leq c|p - q|, \quad (102)$$

so H is globally c -Lipschitz. If $|v| > c$, taking $p = tv/|v|$ gives

$$p \cdot v - H(p) \geq t(|v| - c) - H(0) \longrightarrow +\infty,$$

and hence

$$H^*(v) = +\infty, \quad |v| > c. \quad (103)$$

It remains to determine the boundary value. Let $|v_0| = c$ and $v_s = sv_0$, $0 < s < 1$. From Eq. (101),

$$H^*(v_s) = -cQ \sqrt{1 - s^2} \longrightarrow 0.$$

Lower semicontinuity gives $H^*(v_0) \leq 0$. Since $H^*(0) = -cQ$, convexity gives

$$H^*(v_s) \leq sH^*(v_0) + (1 - s)H^*(0),$$

and therefore $H^*(v_0) \geq 0$ as $s \rightarrow 1^-$. Thus

$$H^*(v) = \begin{cases} -cQ \sqrt{1 - |v|^2/c^2}, & |v| \leq c, \\ +\infty, & |v| > c. \end{cases} \quad (104)$$

A finite convex function on $\mathbb{R}^3$ is closed, hence $H = H^{**}$. Consequently

$$H(p) = \sup_{|v| \leq c} \left[p \cdot v + cQ \sqrt{1 - \frac{|v|^2}{c^2}} \right] = c \sqrt{|p|^2 + Q^2}, \quad (105)$$

the support function of the Euclidean unit half-ball. $\square$

The theorem requires no independent isotropy assumption on H , no strict convexity, and no Hessian nondegeneracy. Its additional input is global: the canonical representation must cover the complete massive MS state manifold.

The rest value $H(0) = cQ$ is a conclusion, not an additional normalization hypothesis.

Two complementary results replace this global hypothesis by different regularity assumptions.

Theorem 3.3 (Radial rigidity). *Let $H \in C^2(\mathbb{R}^3)$ be radial,*

$$H(p) = h(r), \quad r = |p|,$$

where $c, Q > 0$. If Eqs. (88) and (89) hold globally, then either

$$H \equiv cQ \quad (106)$$

or

$$H(p) = c \sqrt{|p|^2 + Q^2}. \quad (107)$$

Proof. Set $e = h/c$. Radial C^2 regularity gives $e'(0) = 0$. The master equation and positive branch at the origin then give $e(0) = Q$. The radial equation becomes

$$e'^2 + \frac{(e - re')^2}{Q^2} = 1, \quad e - re' > 0. \quad (108)$$

There is therefore a C^1 function $\theta \in (-\pi/2, \pi/2)$ such that

$$e' = \sin \theta, \quad \frac{e - re'}{Q} = \cos \theta, \quad \theta(0) = 0.$$

Differentiation gives

$$(r \cos \theta - Q \sin \theta)\theta' = 0. \quad (109)$$

Let $U = \{r > 0 : \theta'(r) \neq 0\}$. On every connected component of U ,

$$\theta(r) = \arctan(r/Q), \quad \theta'(r) = \frac{Q}{r^2 + Q^2} > 0.$$

Continuity of θ' excludes any finite endpoint in $(0, \infty)$ of such a component. Hence either $U = \emptyset$ or $U = (0, \infty)$. In the first case $\theta \equiv 0$, so $H \equiv cQ$. In the second,

$$e'(r) = \frac{r}{\sqrt{r^2 + Q^2}},$$

and $e(0) = Q$ gives $e(r) = \sqrt{r^2 + Q^2}$. $\square$

Theorem 3.4 (Differential rigidity). *Let $H \in C^2(\mathbb{R}^3)$ satisfy Eqs. (88) and (89) for $c, Q > 0$. If*

$$\det \nabla^2 H \neq 0 \quad (110)$$

on a dense subset of $\mathbb{R}^3$, then

$$H(p) = c \sqrt{|p|^2 + Q^2}. \quad (111)$$

Proof. Set

$$G := H - p \cdot \nabla H.$$

Then $\nabla G = -\nabla^2 H p$. Differentiating Eq. (88) yields

$$\nabla^2 H \left(\nabla H - \frac{G}{Q^2} p \right) = 0. \quad (112)$$

On the dense set where the Hessian is invertible,

$$\nabla H = \frac{G}{Q^2} p.$$

Continuity extends this equality to all p . Substitution in the master equation and the condition $G > 0$ give

$$G = \frac{cQ^2}{\sqrt{|p|^2 + Q^2}}, \quad \nabla H = \frac{cp}{\sqrt{|p|^2 + Q^2}}. \quad (113)$$

Finally, $H = G + p \cdot \nabla H = G(1 + |p|^2/Q^2) = c \sqrt{|p|^2 + Q^2}$, which proves the claim without a separate rest-value assumption. $\square$

The extra hypotheses cannot simply be omitted. Let R be an orthogonal projection and define

$$H_R(p) = c \sqrt{Q^2 + |Rp|^2}. \quad (114)$$

Then

$$\nabla H_R = \frac{cRp}{\sqrt{Q^2 + |Rp|^2}}, \quad H_R - p \cdot \nabla H_R = \frac{cQ^2}{\sqrt{Q^2 + |Rp|^2}}, \quad (115)$$

so Eq. (88) holds exactly. If $R \neq I$, however,

$$\nabla H_R(\mathbb{R}^3) \subset \text{Ran } R, \quad \ker R \subseteq \ker \nabla^2 H_R,$$

and both representation completeness and dense Hessian nondegeneracy fail. These MS-invisible canonical directions are analyzed in Appendix C; no gauge interpretation of them is assumed.

We now return to the nondegenerate branch,

$$H_Q(p) = c \sqrt{|p|^2 + Q^2}. \quad (116)$$

Its canonical MS state is

$$D_H = \frac{p}{\sqrt{|p|^2 + Q^2}}, \quad S_H = \frac{Q}{\sqrt{|p|^2 + Q^2}}. \quad (117)$$

Writing

$$E := H_Q(p), \quad P := \left(\frac{E}{c}, p\right), \quad \Pi := P^\flat = \left(-\frac{E}{c}, p\right), \quad (118)$$

the projective map satisfies

$$P = Q\Phi(D_H, S_H). \quad (119)$$

Therefore

$$\eta(P, P) = -Q^2, \quad P^0 > 0. \quad (120)$$

With

$$m := \frac{Q}{c}, \quad (121)$$

this is

$$E^2 - c^2|p|^2 = m^2 c^4. \quad (122)$$

The scalar bridge simultaneously reduces to the standard action relation

$$\Pi(dX) = -mc^2 d\tau. \quad (123)$$

The compact boundary also admits a finite-momentum null completion. At fixed $p \neq 0$,

$$\lim_{Q \rightarrow 0^+} H_Q(p) = c|p|, \quad (D_H, S_H) \longrightarrow \left(\frac{p}{|p|}, 0\right). \quad (124)$$

Thus

$$H_0(p) = c|p|, \quad p \neq 0, \quad (125)$$

with $|v| = c$ and vanishing scalar MS component. The convex-dual interpretation of this limit is given in Appendix B.

A nonzero null momentum is naturally a homogeneous lift of the projective boundary:

$$P = \varrho(1, D), \quad \varrho > 0, \quad |D| = 1. \quad (126)$$

Then

$$\eta(P, P) = 0. \quad (127)$$

In the massive interior,

$$P = \frac{Q}{S}(1, D), \quad \varrho = \frac{Q}{S}, \quad (128)$$

whereas at $S = 0$ the positive null scale ϱ is independent of the vanished massive normalization Q .

The two approaches to the projective boundary are therefore distinct:

$$\begin{aligned} Q > 0, S \rightarrow 0^+ &\implies |p| \rightarrow \infty && \text{on a fixed massive shell,} \\ Q \rightarrow 0^+, p \neq 0 \text{ fixed} &\implies S \rightarrow 0 && \text{with finite null momentum.} \end{aligned} \quad (129)$$

The structure selected in this section can finally be summarized by the two radial realizations

$$\left(\frac{dx}{c}, d\tau\right) = dt(D, S), \quad (d, q) = Q(D, S), \quad (130)$$

together with the scalar incidence relation

$$q dt = Q d\tau = -\frac{1}{c}\Pi(dX).$$

Canonical representation consistency requires both radial realizations to carry the same normalized MS state. Under complete representation or dense Hessian nondegeneracy, this requirement selects the standard relativistic free Hamiltonian. In the radial C^2 case the same conclusion holds after excluding the constant branch.

No observer transformation law has been used in obtaining this result. The next section begins with the dynamically selected massive shell

$$\eta(P, P) = -Q^2, \quad P^0 > 0,$$

and determines its maximal connected covariance group and the conditions under which that covariance is realized by inertial observers.

4 Lorentz and Poincaré Structure

Sections 2 and 3 produced a nested sequence of structures without assuming a Lorentz transformation law between observers. First, the bounded reversible decomposition-comparison principle, together with spatial rotational covariance and axis-orientation independence, selected the Euclidean MS shell up to one positive relative calibration of the motion and staticity channels. The subsequent operational radial realization then induced the quadratic form

$$\eta = \text{diag}(-1, +1, +1, +1), \quad (131)$$

while canonical representation consistency and Hamiltonian rigidity selected the positive massive shell

$$\mathcal{H}_Q^+ := \{P \in \mathbb{R}^4 : \eta(P, P) = -Q^2, P^0 > 0\}, \quad Q > 0. \quad (132)$$

Here

$$P = \left(\frac{E}{c}, p\right), \quad \Pi := P^\flat = \left(-\frac{E}{c}, p\right). \quad (133)$$

We now ask how two inertial operational descriptions of the same process can be related. No absolute observer velocity is introduced. The Lorentz and Poincaré groups obtained below are standard structures of relativistic kinematics [2, 3, 11]; the issue here is their place in the preceding reconstruction.

4.1 Translation duality and Lorentz covariance

For the observer comparison we additionally assume a common affine event structure, with additive translation increments whose restrictions to pure inertial segments are the increments constructed in Sec. 2. This is not a consequence of the barycentric cone extension in Appendix A. Let

$$\mathfrak{t} \simeq \mathbb{R}^4 \quad (134)$$

denote the real vector space generated by operational translation increments,

$$dX = (c \, dt, dx) \in \mathfrak{t}. \quad (135)$$

The canonical generator belongs naturally to the dual space,

$$\Pi \in \mathfrak{t}^*, \quad \Pi(dX) = p \cdot dx - E \, dt. \quad (136)$$

Thus the generator–translation pairing is defined before any use is made of a Lorentz transformation law.

We first record the maximal linear covariance of the massive shell. Here $O^\uparrow(1, 3)$ denotes the subgroup of $O(1, 3)$ preserving the chosen future time orientation, while $SO^+(1, 3)$ denotes its identity component.

Theorem 4.1 (Linear automorphisms of the positive massive shell). *The real-linear automorphism group of $\mathcal{H}_Q^+$ is*

$$\text{Aut}_{\text{lin}}(\mathcal{H}_Q^+) = O^\uparrow(1, 3), \quad (137)$$

and its identity component is

$$\text{Aut}_{\text{lin}}^0(\mathcal{H}_Q^+) = SO^+(1, 3). \quad (138)$$

Proof. Let $A \in \text{GL}(4, \mathbb{R})$ map $\mathcal{H}_Q^+$ onto itself and set

$$C := A^T \eta A - \eta.$$

Using the parametrization

$$P(p) = \left(\sqrt{Q^2 + |p|^2}, p \right),$$

shell preservation implies

$$P(p)^T C P(p) = 0 \quad \text{for all } p \in \mathbb{R}^3. \quad (139)$$

At $p = 0$ this gives $C_{00} = 0$. Differentiating with respect to p_i at the origin gives $C_{0i} = 0$. Equation (139) then reduces to

$$C_{ij} p^i p^j = 0 \quad \text{for all } p,$$

so $C_{ij} = 0$ and therefore

$$A^T \eta A = \eta.$$

Because A preserves the positive sheet, $A \in O^\uparrow(1, 3)$. The converse is immediate, and the identity component is $SO^+(1, 3)$. $\square$

Now consider two inertial operational representations A and B . Their translation increments are related by a continuous invertible map

$$M_{BA} : \mathfrak{t} \longrightarrow \mathfrak{t}. \quad (140)$$

Preservation of process concatenation gives

$$M_{BA}(X + Y) = M_{BA}X + M_{BA}Y. \quad (141)$$

A continuous additive map between finite-dimensional real vector spaces is real linear. This linearity follows from process composition and does not assume Lorentz covariance.

We require the canonical pairing to be observer independent as a bilinear pairing on the entire translation space: for each corresponding pair of generators, and for every $dX_A \in \mathfrak{t}$,

$$\Pi_B(M_{BA}dX_A) = \Pi_A(dX_A). \quad (142)$$

This is a covariance assumption for the full translation response, stronger than equality evaluated only on the generator’s own trajectory. Hence

$$\Pi_B = M_{BA}^{-T} \Pi_A. \quad (143)$$

Using $P = \eta^\sharp(\Pi)$, the associated generator-vector transformation is

$$P_B = A_{BA}P_A, \quad A_{BA} := \eta M_{BA}^{-T} \eta. \quad (144)$$

Suppose the two observers describe the same massive species and therefore the same intrinsic scale Q . Then A_{BA} maps $\mathcal{H}_Q^+$ onto itself. By Theorem 4.1,

$$A_{BA}^T \eta A_{BA} = \eta.$$

For a Lorentz matrix,

$$A_{BA}^{-T} = \eta A_{BA} \eta.$$

Equation (144) also gives $M_{BA}^{-T} = \eta A_{BA} \eta$, and therefore

$$M_{BA} = A_{BA}. \quad (145)$$

Thus the same Lorentz matrix acts on translation increments and generator vectors:

$$dX_B = \Lambda_{BA} dX_A, \quad P_B = \Lambda_{BA} P_A, \quad \Lambda_{BA} \in O^\uparrow(1, 3). \quad (146)$$

For observer changes continuously connected to the identity,

$$\Lambda_{BA} \in \text{SO}^+(1, 3). \quad (147)$$

This is the first stage at which transformations between observer-dependent durations enter. Each observer has already assigned its own radial duration through the operational construction of Sec. 2; Lorentz covariance determines the relation between those assignments.

4.2 Observer realization, universal scale, and affine completion

Let G_{obs} be a connected Lie group of inertial observer changes and let

$$\Lambda : G_{\text{obs}} \longrightarrow \text{SO}^+(1, 3) \quad (148)$$

be the smooth representation obtained above. The shell argument constrains the observer image but does not by itself imply that every Lorentz transformation is physically realized.

Corollary 4.2 (Full connected Lorentz realization). *Suppose that $\Lambda(G_{\text{obs}})$ contains the full spatial rotation group $\text{SO}(3)$ and acts transitively on $\mathcal{H}_Q^+$. Then*

$$\Lambda(G_{\text{obs}}) = \text{SO}^+(1, 3). \quad (149)$$

If Λ is faithful, then $G_{\text{obs}} \simeq \text{SO}^+(1, 3)$.

Proof. Set $H_{\text{obs}} = \Lambda(G_{\text{obs}})$, and let $P_0 = (Q, 0)$. Its stabilizer in $\text{SO}^+(1, 3)$ is the spatial rotation group $K = \text{SO}(3)$, which is contained in H_{obs} . For any $g \in \text{SO}^+(1, 3)$, transitivity gives $h \in H_{\text{obs}}$ such that $hP_0 = gP_0$. Hence $h^{-1}g \in K \subset H_{\text{obs}}$, and therefore $g \in H_{\text{obs}}$. Thus $H_{\text{obs}} = \text{SO}^+(1, 3)$. If Λ is also faithful, it is a bijective Lie-group homomorphism, and hence a Lie-group isomorphism. $\square$

The transitivity assumption is equivalent to comoving-frame completeness: every massive inertial state admits an observer for which its generator is represented by $P_0 = (Q, 0)$. Full Lorentz realization therefore requires more than invariance of the shell itself.

We next address the conversion scale c . Let two massive species A and B be described using common physical units for spatial displacement and observer duration, while allowing a priori distinct conversion constants c_A and c_B . Suppose their generators transform under the same observer action and that this action contains a transformation mixing temporal and spatial directions.

Proposition 4.3 (Universality of the conversion scale). *If the massive shell of each species is invariant under the common observer action, then*

$$c_A = c_B. \quad (150)$$

Proof. Choose an arbitrary reference constant $c_0 > 0$ and write

$$Y = \left(\frac{E}{c_0}, p \right).$$

For species $s \in \{A, B\}$, its shell is

$$Y^T G_s Y = -Q_s^2, \quad G_s = \text{diag} \left(-\frac{c_0^2}{c_s^2}, 1, 1, 1 \right). \quad (151)$$

By the shell-rigidity argument of Theorem 4.1, invariance of each nonzero massive hyperboloid implies

$$B_g^T G_s B_g = G_s$$

for every common observer transformation B_g . It follows that B_g commutes with

$$G_A^{-1} G_B = \text{diag} \left(\frac{c_A^2}{c_B^2}, 1, 1, 1 \right). \quad (152)$$

If $c_A \neq c_B$, the one-dimensional temporal eigenspace and the three-dimensional spatial eigenspace are distinct and must be preserved separately by every B_g . No common transformation could then mix temporal and spatial directions. Hence $c_A = c_B$. $\square$

Thus a common nontrivial relativistic observer structure makes the local conversion scale species independent. The same c defines the null boundary preserved by the resulting Lorentz group. Identifying this causal scale with the propagation speed of a particular massless field requires the additional physical statement that the field realizes this null sector.

Once the event set is organized as an affine space modeled on the translation vector space $\mathfrak{t}$, freedom in the choice of origin gives

$$X' = \Lambda X + a, \quad a \in \mathfrak{t}. \quad (153)$$

The maximal connected affine covariance group is therefore

$$\mathcal{P}_+^\uparrow = \mathfrak{t} \rtimes \text{SO}^+(1, 3) \simeq \mathbb{R}^4 \rtimes \text{SO}^+(1, 3), \quad (154)$$

the proper orthochronous Poincaré group.

Finally, the linear Lorentz action induces the familiar nonlinear action on the bounded MS variables. Put

$$Y = (1, D)$$

and, for $\Lambda \in \text{SO}^+(1, 3)$, define

$$a(\Lambda, D) := \Lambda^0_0 + \Lambda^0_j D^j. \quad (155)$$

For every future causal state $a(\Lambda, D) > 0$. Writing

$$\Lambda(1, D) = a(\Lambda, D)(1, D')$$

and using preservation of $\eta((1, D), (1, D)) = -S^2$ gives

$$D'^i = \frac{\Lambda^i_0 + \Lambda^i_j D^j}{\Lambda^0_0 + \Lambda^0_j D^j}, \quad (156)$$

$$S' = \frac{S}{\Lambda^0_0 + \Lambda^0_j D^j}. \quad (157)$$

For $S > 0$, equivalently,

$$(D', S') = \Phi^{-1} \circ \Lambda \circ \Phi(D, S). \quad (158)$$

With $D = v/c$, Eq. (156) is the standard relativistic velocity transformation law.

The same action extends continuously to the null boundary. For

$$P = \varrho(1, D), \quad \varrho > 0, \quad |D| = 1, \quad (159)$$

one has

$$P' = \varrho'(1, D'), \quad \varrho' = a(\Lambda, D)\varrho. \quad (160)$$

Thus Lorentz transformations act projectively on the compact causal direction and homogeneously on the positive null generator scale.

The result of this section is therefore a separation between maximal covariance and physical realization. The massive shell fixes $\text{SO}^+(1, 3)$ as the maximal connected linear covariance group, whereas the realization of the full group by inertial observers requires comoving-frame completeness and unrestricted spatial reorientation. Observer duration is defined locally before this step; Lorentz covariance determines how the durations and spatial displacements assigned by different inertial observers are related.

5 Discussion and Outlook

We have formulated relativistic free-particle dynamics in terms of a bounded motion–staticity state

$$(D, S), \quad |D|^2 + S^2 = 1, \quad S > 0,$$

and required the same state to admit compatible operational and canonical representations. In contrast with taking the Euclidean MS shell as an independent geometric ansatz, Sec. 2 derives its signed completion, up to one positive relative channel calibration, from continuous reversible linear decomposition comparison with a bounded spanning pure-staticity orbit, spatial rotational covariance, and independence of the arbitrary orientation of a spatial axis; positive scalar orientation then selects the

physical branch $S > 0$. The boundedness condition is not intended as an independent maximum-speed postulate. It expresses the prior physical meaning assigned to staticity: motion and staticity are complementary components of a fixed normalized relational state, whereas overall magnitude is carried separately by radial variables. Allowing unbounded motion gain inside the normalized decomposition sector would destroy that complementarity by giving the motion coordinate an additional extensive scale. In this sense, bounded motion–staticity composition is the upstream structural origin, within the present reconstruction, of the finite motion fraction that is later mapped operationally to the limiting speed c . The familiar proper-time relation, future unit hyperboloid, relativistic mass shell, Lorentz group, Poincaré group, and Legendre–Fenchel duality are standard structures [9, 10, 11, 15, 16]. The main content of the present work is the two-stage rigidity problem: first the selection of the compact MS geometry from the decomposition structure, and then the selection of the standard free Hamiltonian from mutually consistent operational and canonical representations of that state.

The temporal assumptions should be stated precisely. The instantaneous compact state does not generate a time parameter by itself: a free inertial state can remain constant along an entire history. The construction therefore assumes a positive additive scalar process accumulation $d\chi$ and a spatial process increment. Their observer-dependent radial realization defines dt , after which the induced Lorentzian relation permits the identification $d\tau = d\chi$. Thus coordinate duration is not introduced as an external Newtonian parameter, but neither temporal ordering nor the existence of intrinsic process accumulation is derived from $|D|^2 + S^2 = 1$ alone.

The central dynamical bridge is the identification of the same MS scalar component in the operational and canonical representations,

$$q = Q \frac{d\tau}{dt} = \frac{H - p \cdot \nabla H}{c}, \quad q dt = Q d\tau = -\frac{1}{c} \Pi(dX). \quad (161)$$

When a regular Lagrangian description exists, Legendre duality gives the additional equivalent representation

$$q = -\frac{L}{c} = \frac{H - p \cdot v}{c}.$$

The individual proper-time and Legendre identities are standard. The additional requirement is that the operational and canonical quantities describe the same scalar component of one compact state while $D = v/c = \nabla H/c$.

Combining this common representation with the compact normalization produces the Hamiltonian consistency equation

$$\frac{|\nabla H|^2}{c^2} + \frac{(H - p \cdot \nabla H)^2}{c^2 Q^2} = 1, \quad H - p \cdot \nabla H > 0. \quad (162)$$

The strongest inverse result shows that a finite convex C^1 Hamiltonian satisfying Eq. (162) globally and representing the complete massive MS manifold is necessarily

$$H(p) = c \sqrt{|p|^2 + Q^2}. \quad (163)$$

No independent Hamiltonian isotropy, strict convexity, or Hessian nondegeneracy is required, and $H(0) = cQ$ follows. The differential rigidity theorem and the radial theorem with its constant branch excluded provide complementary sufficient conditions. Conversely, the projection family of Sec. 3 and Appendix C shows that the pointwise consistency equation alone permits lower-dimensional canonical realizations. The additional hypotheses therefore remove genuine representation degeneracies rather than merely technical irregularities.

The forward convex construction gives a complementary diagnostic. If the decomposition-rigidity hypotheses of Sec. 2 are relaxed and an arbitrary compact convex state body K is supplied, then

$$H_{K,Q}(p) = c h_K(p, Q),$$

so different state geometries generally give different massive dispersion relations. Thus neither compactness nor Hamiltonian representation consistency alone selects the Euclidean shell. Likewise, the conventional Newtonian quadratic limit constrains the local second-order curvature at the static state but does not determine the global state geometry for a general compact body. In the present construction the logical order is instead upstream: the Euclidean MS shell is selected by the bounded decomposition-comparison structure of Sec. 2, and Sec. 3 then asks which Hamiltonians can represent that already selected state manifold completely.

For the Euclidean state body the rigid Hamiltonian yields

$$P = \left(\frac{E}{c}, p \right), \quad \eta(P, P) = -Q^2,$$

and, with $Q = mc$, the standard massive dispersion relation follows. The projective boundary $S = 0$ represents null directions. Two limits must nevertheless be distinguished: $S \rightarrow 0^+$ at fixed $Q > 0$ is an infinite-momentum limit on one massive shell, whereas $Q \rightarrow 0^+$ at fixed nonzero momentum produces the finite null dispersion $E = c|p|$. This construction identifies a free null sector, not the dynamics of any particular massless field.

The symmetry result has a similarly restricted meaning. The positive mass shell has $\text{SO}^+(1, 3)$ as its maximal connected linear covariance group, and translation-origin freedom gives the proper orthochronous Poincaré group. A physical family of inertial observers need not realize the entire Lorentz group unless additional completeness assumptions are made. In the present framework the Lorentz transformation does not create observer duration; it relates the different radial durations and spatial increments assigned to the same process by different inertial observers.

Several assumptions therefore remain explicit. We assume a three-dimensional Euclidean spatial readout and its rotational organization, a primitive vector–scalar motion–staticity splitting, continuous reversible linear decomposition comparison with a bounded spanning pure-staticity orbit, spatial rotational covariance, independence of the arbitrary orientation assigned to each spatial axis, positive additive intrinsic accumulation along massive processes, and the first-moment conditions used to obtain the linear operational readout. The signed comparison completion used in the decomposition theorem is auxiliary; the physical massive branch retains positive scalar orientation. The construction does not derive spatial dimension, the vector–scalar split, temporal ordering, or the comparison principles themselves from a structureless set of events. Its narrower claim is that these inputs first rigidly select the compact Euclidean MS geometry up to channel calibration and then, together with common operational–canonical representation and suitable completeness or nondegeneracy conditions, rigidly select the standard massive free Hamiltonian and its Lorentz–Poincaré covariance.

The generator–translation pairing

$$\Pi(dX) = p \cdot dx - E dt$$

is particularly useful for extensions because it does not require one to regard momentum primarily as a coordinate four-vector. In the present flat problem the translation space is global and the metric identifies generator vectors and covectors. In a curved geometry there is generally no canonical global translation group or path-independent addition of local momenta. A natural extension is therefore to investigate local generator–displacement response before introducing any global momentum notion.

Interactions raise a separate issue. The free-particle relation

$$-\Pi(dX) = cQ d\tau$$

does not determine the generator of an interacting or composite system, whose energy and momentum include interaction contributions. Spin, internal charges, fields, quantum systems, and gravitation likewise require structures absent from the present analysis.

The result should consequently be read as a reconstruction and rigidity statement for classical relativistic free-particle dynamics. The logical order is two-stage. Bounded decomposition comparison first selects the compact MS state geometry up to channel calibration. That state is then required to admit compatible operational, canonical-generator, and relational observer descriptions. Together with the stated completeness or nondegeneracy hypotheses, the second stage selects the standard relativistic massive Hamiltonian and the connected Lorentz–Poincaré covariance structure.

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A Operational Cone and Linear Extension

This appendix gives the linear-extension argument used in Sec. 2. After the channel calibration selected by Theorem 2.2 and restriction to the positive scalar branch, the normalized massive state manifold is

$$\mathcal{S} = \{(D, S) \in \mathbb{R}^3 \oplus \mathbb{R} : |D|^2 + S^2 = 1, S > 0\}. \quad (\text{A1})$$

No linear structure is imposed directly on $\mathcal{S}$. Positive scaling is introduced only after passing to unnormalized process or ensemble data.

Let

$$V := \mathbb{R}^3 \oplus \mathbb{R}.$$

For a finite positive measure μ on $\mathcal{S}$, define its unnormalized barycentric state by

$$Y_\mu := \int_{\mathcal{S}} X d\mu(X), \quad X = (D, S), \quad (\text{A2})$$

and let

$$C := \{Y_\mu : \mu \text{ is a finite positive measure on } \mathcal{S}\}. \quad (\text{A3})$$

Equivalently, C is the positive cone generated by $\mathcal{S}$.

Suppose a preparation μ has first-moment operational readout

$$M(\mu) \in W, \quad W := \mathbb{R}_{\text{disp}}^3 \oplus \mathbb{R}_{\mathcal{X}}. \quad (\text{A4})$$

We assume barycentric sufficiency,

$$Y_\mu = Y_\nu \implies M(\mu) = M(\nu), \quad (\text{A5})$$

together with additivity under combination of preparations and positive homogeneity under common positive rescaling. Hence

$$R(Y_\mu) := M(\mu) \quad (\text{A6})$$

defines an additive, positively homogeneous map $R : C \rightarrow W$.

Lemma A.1. *The cone C generates the whole ambient space:*

$$C - C = V. \quad (\text{A7})$$

Proof. The static state

$$X_0 = (0, 1)$$

belongs to $\mathcal{S}$. For any $a \in (0, 1)$ and standard basis vectors e_i , the states

$$X_i^\pm = (\pm ae_i, \sqrt{1 - a^2})$$

also belong to $\mathcal{S}$, and

$$X_i^+ - X_i^- = (2ae_i, 0).$$

Thus $\mathcal{S}$, and hence C , spans $\mathbb{R}^3 \oplus \mathbb{R}$. Since C is a convex cone, every real linear combination can be separated into its positive and negative parts, proving Eq. (A7). $\square$

Proposition A.2 (Unique linear extension). *There exists a unique real-linear map*

$$\widetilde{R} : V \longrightarrow W \quad (\text{A8})$$

whose restriction to C is R .

Proof. By Lemma A.1, write $Z = Y_+ - Y_-$ with $Y_{\pm} \in C$ and define

$$\widetilde{R}(Z) := R(Y_+) - R(Y_-). \quad (\text{A9})$$

If also $Z = Y'_+ - Y'_-$, then $Y_+ + Y'_- = Y'_+ + Y_-$. Additivity of R therefore gives

$$R(Y_+) - R(Y_-) = R(Y'_+) - R(Y'_-),$$

so the definition is independent of the decomposition. Additivity and positive homogeneity of R imply additivity and real homogeneity of $\widetilde{R}$. Any other linear extension must have the same value on every difference $Y_+ - Y_-$, proving uniqueness. $\square$

We next impose spatial covariance. The rotation group acts on both spaces by

$$R \oplus 1, \quad R \in \text{SO}(3),$$

and we require

$$\widetilde{R}(RD, S) = (R \oplus 1)\widetilde{R}(D, S). \quad (\text{A10})$$

Proposition A.3 (Rotational form of the readout). *Every real-linear map satisfying Eq. (A10) has the form*

$$\widetilde{R}(D, S) = (\alpha D, \beta S) \quad (\text{A11})$$

for constants $\alpha, \beta \in \mathbb{R}$.

Proof. Write

$$\widetilde{R}(D, S) = (AD + bS, c \cdot D + \beta S).$$

Equivariance gives $AR = RA$ for every $R \in \text{SO}(3)$, hence $A = \alpha I$. It also requires $Rb = b$ and $R^T c = c$ for every rotation, so $b = c = 0$. $\square$

The operational assumptions of Sec. 2 select $\alpha, \beta > 0$. For a pure process ray

$$Y = \lambda(D, S), \quad \lambda > 0,$$

the readout is therefore

$$(dx, d\chi) = \lambda(\alpha D, \beta S). \quad (\text{A12})$$

Defining

$$c := \frac{\alpha}{\beta}, \quad dt := \beta \lambda \quad (\text{A13})$$

gives

$$\left(\frac{dx}{c}, d\chi \right) = dt(D, S). \quad (\text{A14})$$

Equation (A14) is a calibration of the two irreducible readout channels, not part of the abstract extension theorem. In particular, the observer duration t is introduced only after the additive cone map has been extended and its rotational form has been determined.

The extension concerns $(dx, d\chi)$, not an additive duration on arbitrary barycentric preparations. Indeed, for $Y_\mu = \int X d\mu$, the total weight $\mu(S)$ need not be determined by Y_μ . Retaining the additive quantity $T_\mu = \beta \mu(S)$ would require this additional preparation datum. The pure-ray duration must therefore be distinguished from the duration of a concatenated history.

B Convex-Dual Examples and Newtonian Rest Geometry

This appendix gives explicit examples for the forward convex-dual construction used in Sec. 3. The underlying Legendre–Fenchel and support-function results are standard [15, 16].

For $r > 1$, define

$$K_r^+ := \{(D, S) : |D|^r + S^r \leq 1, S \geq 0\}. \quad (\text{B15})$$

Its vector projection is the Euclidean unit ball and its upper boundary is

$$\sigma_r(D) = (1 - |D|^r)^{1/r}, \quad |D| \leq 1. \quad (\text{B16})$$

Let

$$r' := \frac{r}{r-1}, \quad \frac{1}{r} + \frac{1}{r'} = 1. \quad (\text{B17})$$

Proposition B.1. For $Q \geq 0$,

$$h_{K_r^+}(p, Q) = (|p|^{r'} + Q^{r'})^{1/r'}, \quad (\text{B18})$$

and therefore

$$H_{r,Q}(p) = c(|p|^{r'} + Q^{r'})^{1/r'}. \quad (\text{B19})$$

Proof. For $(D, S) \in K_r^+$, Cauchy–Schwarz followed by Hölder’s inequality gives

$$\begin{aligned} p \cdot D + QS &\leq |p||D| + QS \\ &\leq (|p|^{r'} + Q^{r'})^{1/r'} (|D|^r + S^r)^{1/r} \\ &\leq (|p|^{r'} + Q^{r'})^{1/r'}. \end{aligned} \quad (\text{B20})$$

For $p \neq 0$ and $Q > 0$, equality is attained by taking

$$D = \frac{|p|^{r'-1} p}{(|p|^{r'} + Q^{r'})^{1/r}}, \quad S = \frac{Q^{r'-1}}{(|p|^{r'} + Q^{r'})^{1/r}}.$$

The cases $p = 0$ or $Q = 0$ follow directly or by continuity. Equation (B19) then follows from $H_{K,Q} = c h_{K_r^+}(p, Q)$. $\square$

For $Q > 0$, the corresponding extended-real Lagrangian is

$$L_{r,Q}(v) = \begin{cases} -cQ \left[1 - \left(\frac{|v|}{c} \right)^r \right]^{1/r}, & |v| \leq c, \\ +\infty, & |v| > c. \end{cases} \quad (\text{B21})$$

The boundary $|v| = c$ belongs to the closed convex extension; for fixed $Q > 0$ it is not a finite-momentum massive state.

The Euclidean state body is the case $r = r' = 2$, for which

$$H_{2,Q}(p) = c \sqrt{|p|^2 + Q^2}.$$

For $p := |p| \ll Q$,

$$H_{r,Q}(p) = cQ + \frac{cQ}{r'} \left(\frac{p}{Q} \right)^{r'} + O \left[cQ \left(\frac{p}{Q} \right)^{2r'} \right]. \quad (\text{B22})$$

With $m = Q/c$, the standard Newtonian behavior is

$$H(p) = mc^2 + \frac{p^2}{2m} + o(p^2). \quad (\text{B23})$$

Within the K_r^+ family this requires $r' = 2$, and hence

$$r = 2. \quad (\text{B24})$$

For $r > 2$, one has $1 < r' < 2$ and the leading correction is lower than quadratic. For $1 < r < 2$, one has $r' > 2$ and the quadratic kinetic coefficient vanishes. Thus the conventional nondegenerate Newtonian rest behavior selects the Euclidean member of this particular family.

This conclusion is only family-specific. For a general smooth compact state body, the Newtonian limit constrains only the local curvature of the upper boundary.

Proposition B.2 (Newtonian rest-curvature condition). *Suppose the upper boundary of a compact state body is*

$$S = \sigma(D)$$

near $D = 0$, with $\sigma \in C^2$,

$$\sigma(0) = 1, \quad \nabla \sigma(0) = 0.$$

For

$$L_Q(v) = -cQ \sigma\left(\frac{v}{c}\right), \quad (\text{B25})$$

the isotropic Newtonian expansion

$$L_Q(v) = -mc^2 + \frac{1}{2}m|v|^2 + o(|v|^2), \quad m = \frac{Q}{c}, \quad (\text{B26})$$

holds if and only if

$$D^2\sigma(0) = -I. \quad (\text{B27})$$

Proof. Taylor expansion gives

$$\sigma(D) = 1 + \frac{1}{2}D^T D^2\sigma(0)D + o(|D|^2).$$

Substitution of $D = v/c$ gives

$$L_Q(v) = -cQ - \frac{Q}{2c}v^T D^2\sigma(0)v + o(|v|^2). \quad (\text{B28})$$

Since $m = Q/c$, comparison with Eq. (B26) is equivalent to $D^2\sigma(0) = -I$. $\square$

For a rotationally invariant state body, $D^2\sigma(0) = -\lambda I$; the Newtonian normalization fixes $\lambda = 1$. This is only a local second-order condition. It does not imply globally that

$$\sigma(D) = \sqrt{1 - |D|^2}.$$

Infinitely many compact convex bodies can agree with the Euclidean half-ball to second order at the static point and differ elsewhere.

The scale-free sector contains still less information. Setting $Q = 0$ in Eq. (B19) gives

$$H_{r,0}(p) = c|p| \quad (\text{B29})$$

for every $r > 1$. More generally, if Ω is the vector projection of a compact state body K , then

$$H_{K,0}(p) = c h_K(p, 0) = c h_\Omega(p). \quad (\text{B30})$$

Hence the $Q = 0$ dispersion probes only the projected velocity geometry, whereas the massive sector $Q > 0$ also probes the scalar profile of the state body.

For the Euclidean body,

$$\sigma_2(D) = \sqrt{1 - |D|^2}, \quad D^2 \sigma_2(0) = -I,$$

and

$$H_{2,Q}(p) = c \sqrt{|p|^2 + Q^2}.$$

Thus the Newtonian rest limit distinguishes the Euclidean member within the explicit K_r^+ family, but does not by itself establish global Euclidean state geometry in the general case. This limitation is logically independent of Theorem 2.2, which selects the Euclidean MS shell from the stronger bounded decomposition-comparison structure rather than from the Newtonian limit.

C Degenerate Solutions and MS-Invisible Canonical Directions

The pointwise MS consistency equation admits smooth convex solutions that fail both representation completeness and Legendre nondegeneracy. This appendix analyzes the projection family introduced in Sec. 3.

Let

$$R^2 = R, \quad R^T = R \quad (\text{C31})$$

be an orthogonal projection on $\mathbb{R}^3$, with

$$k := \text{rank } R.$$

Decompose

$$p = p_{\parallel} + p_{\perp}, \quad p_{\parallel} := Rp, \quad p_{\perp} := (I - R)p. \quad (\text{C32})$$

Define

$$H_R(p) := c \sqrt{Q^2 + |p_{\parallel}|^2}, \quad Q > 0. \quad (\text{C33})$$

Proposition C.1 (Projection solutions). *The Hamiltonian H_R is smooth and convex and satisfies*

$$H_R(0) = cQ \quad (\text{C34})$$

and

$$\frac{|\nabla H_R|^2}{c^2} + \frac{(H_R - p \cdot \nabla H_R)^2}{c^2 Q^2} = 1, \quad H_R - p \cdot \nabla H_R > 0. \quad (\text{C35})$$

Proof. The function $z \mapsto c \sqrt{Q^2 + |z|^2}$ is smooth and convex for $Q > 0$, and H_R is its pullback by the linear map R . Moreover,

$$\nabla H_R = \frac{c p_{\parallel}}{\sqrt{Q^2 + |p_{\parallel}|^2}}, \quad (\text{C36})$$

and, since $p \cdot p_{\parallel} = |p_{\parallel}|^2$,

$$H_R - p \cdot \nabla H_R = \frac{c Q^2}{\sqrt{Q^2 + |p_{\parallel}|^2}}. \quad (\text{C37})$$

Substitution gives Eq. (C35). $\square$

The corresponding MS state is

$$D_R = \frac{p_{\parallel}}{\sqrt{Q^2 + |p_{\parallel}|^2}}, \quad S_R = \frac{Q}{\sqrt{Q^2 + |p_{\parallel}|^2}}. \quad (\text{C38})$$

It depends only on $p_{\parallel}$.

Proposition C.2 (Canonical image). *The canonical state map generated by H_R has image*

$$\Psi_R(\mathbb{R}^3) = \mathcal{S}_R := \{(D, S) \in \mathcal{S} : D \in \text{Ran } R\}, \quad (\text{C39})$$

or equivalently

$$\nabla H_R(\mathbb{R}^3) = \{v \in \text{Ran } R : |v| < c\}. \quad (\text{C40})$$

Hence the image has dimension $k = \text{rank } R$.

Proof. Equation (C36) gives the inclusion

$$\nabla H_R(\mathbb{R}^3) \subset \{v \in \text{Ran } R : |v| < c\}.$$

Conversely, for any $v \in \text{Ran } R$ with $|v| < c$, choose

$$p_{\parallel} = \frac{Qv}{\sqrt{c^2 - |v|^2}}. \quad (\text{C41})$$

Then $\nabla H_R = v$. The scalar component follows from $S = \sqrt{1 - |D|^2}$ on the positive branch. $\square$

Thus $k = 3$ is the representation-complete case within this family; for $k < 3$, the canonical image is a lower-dimensional section of the massive MS manifold.

The differential degeneracy is equally explicit:

$$\nabla^2 H_R = c \left[\frac{R}{\sqrt{Q^2 + |p_{\parallel}|^2}} - \frac{p_{\parallel} \otimes p_{\parallel}}{(Q^2 + |p_{\parallel}|^2)^{3/2}} \right]. \quad (\text{C42})$$

Proposition C.3 (Exact Hessian kernel). *For every p ,*

$$\ker \nabla^2 H_R(p) = \ker R. \quad (\text{C43})$$

In particular, $\text{rank } \nabla^2 H_R = k$ everywhere.

Proof. Every $w \in \ker R$ satisfies $Rw = 0$ and $p_{\parallel} \cdot w = 0$, so Eq. (C42) gives $\nabla^2 H_R w = 0$.

On the other hand, for $z \in \text{Ran } R$,

$$\begin{aligned} z^T \nabla^2 H_R z &= \frac{c}{\sqrt{Q^2 + |p_{\parallel}|^2}} \left[|z|^2 - \frac{(p_{\parallel} \cdot z)^2}{Q^2 + |p_{\parallel}|^2} \right] \\ &\geq \frac{cQ^2}{(Q^2 + |p_{\parallel}|^2)^{3/2}} |z|^2. \end{aligned} \quad (\text{C44})$$

This is positive for nonzero z , proving the claim. $\square$

The degeneracy persists along the affine leaves

$$p + \ker R. \quad (\text{C45})$$

Indeed, for every $w \in \ker R$,

$$H_R(p + w) = H_R(p), \quad \Psi_R(p + w) = \Psi_R(p). \quad (\text{C46})$$

Thus the dependence of both the Hamiltonian and the MS state map factors through

$$\mathbb{R}^3 / \ker R \simeq \text{Ran } R. \quad (\text{C47})$$

On this quotient their functional dependence is the standard lower-dimensional massive square root,

$$H_R = c \sqrt{Q^2 + |p_{\parallel}|^2}.$$

No claim is made that $\ker R$ is a gauge redundancy of the full canonical phase space; the statement is only that these directions are invisible to the assumed MS state representation.

This distinction is also visible in the generator vector. For a general Hamiltonian we introduced

$$P_H = \left(\frac{H}{c}, p \right). \quad (\text{C48})$$

For the projection solution,

$$\eta(P_H, P_H) = -Q^2 + |p_{\perp}|^2. \quad (\text{C49})$$

Thus P_H does not in general lie on the standard massive shell when $R \neq I$.

The projected vector

$$P_R^{\text{vis}} := \left(\frac{H_R}{c}, p_{\parallel} \right) \quad (\text{C50})$$

does satisfy

$$\eta(P_R^{\text{vis}}, P_R^{\text{vis}}) = -Q^2. \quad (\text{C51})$$

This vector records only the generator component visible to the MS state; it is not asserted to replace the full canonical generator of three-dimensional translations.

Consequently, the pointwise master equation constrains the combination

$$\nabla H, \quad H - p \cdot \nabla H,$$

but does not by itself guarantee that every canonical momentum direction is faithfully represented in the MS state. In the projection family,

$$R = I \iff \Psi_R(\mathbb{R}^3) = \mathcal{S} \iff \nabla^2 H_R \text{ is nondegenerate everywhere.} \quad (\text{C52})$$

Equivalently, $R = I$ is the unique member for which the full vector P_H lies on $\eta(P_H, P_H) = -Q^2$ for every canonical momentum.

The projection family therefore gives explicit smooth convex counterexamples to rigidity based on the master equation alone. Representation completeness excludes missing MS motion directions, while dense Hessian nondegeneracy excludes persistent canonical directions that the MS representation does not detect. No classification of all globally degenerate solutions is attempted here.

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